CP115
Computer Programming 1
Semester I
Session 2026/2027
2 hours 30 min

ULTIMATE SPARK

CP115
Pengaturcaraan Komputer 1
Semester I
Sesi 2026/2027
2 jam 30 min

No. Matrik *Matric No.*	**No. Kad Pengenalan** *Identity Card No.*	**Nama** *Name.*
		Class:

(*Isikan maklumat dengan lengkap*)

KOLEJ MATRIKULASI KEDAH
KEDAH MATRICULATION COLLEGE

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
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Untuk Kegunaan Pemeriksa *For Examiner's Use*		
No. Soalan Question No.	Markah *Mark*	
	Pemeriksa *Examiner*	KP / KKP
1 (22)		
2 (25)		
3 (14)		
4 (7)		
5 (8)		
6 (14)		
JUMLAH *TOTAL*		

Kertas soalan ini mengandungi **8** halaman bercetak.
This question paper consists of **8** *printed pages.*
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1

a) Problem-solving typically comprises of five essential steps. Based on the explanation provided, identify **two (2)** steps that accurately correspond to their respective descriptions. [2 marks]

	Description	**Problem Solving Step**
(i)	This step is performed to write the program code based on the designed solution.	
(ii)	This step is performed to understand the problem, such as identifying what information is needed to order food, calculate the total price and display the order details.	

b) During the software execution and testing process, programmers often encounter different categories of programming errors.

(i) Distinguish between a **Logic Error** and a **Runtime Error** in Python. [2 marks]

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(ii) A student wrote a program to compute the average test mark for a class. When the user inputs `0` for the total number of students, the program terminates abruptly with a `ZeroDivisionError`. Identify and explain which step in the 5-step problem-solving process should have prevented this issue. [2 marks]

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_____

c)

(i) Define IPO Model. [2 marks]

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_____

(ii) For the following description, identify either if it is input, process or output. [3 marks]

Comparing humidity levels against threshold conditions to determine the probability of rain.	
User selecting a specific city district on the app map interface.	
Sending a pop-up push notification warning users about severe heatwave conditions.	

d) Analyze the following real-world logistics scenario and complete the **Input-Process-Output (IPO) Analysis**:

(i) A Smart Parcel Import Taxation System calculates import tariffs and surcharges based on package category, declared value (RM):

- **Category 'E' (Electronics):** Base tariff of 10% on declared value.
- **Category 'C' (Clothing):** Base tariff of 5% on declared value.
- **Category 'F' (Food):** Base tariff of 4% on declared value.

.

The system prompts for category code, and declared value, computes and displays the total import tax payable. If an invalid category code is entered, an error message is generated. [6 marks]

Input	

Process	
Output	

(ii) Due to the ongoing heatwave in Malaysia, households have increased their usage of air conditioners. TNB wants to develop a system to estimate the daily electricity cost of using an air conditioner. The air conditioner consumes **1.5 kWh per hour**, and the electricity rate is **RM0.50 per kWh**. The system will prompt the number of hours the air conditioner is used in a day to calculate and display the total cost for that day. [5 marks]

Input	
Process	
Output	

2

a) List and explain all control structure in programming. [3 marks]

b) Petronas plans to implement a system to monitor the gas pressure in its pipeline following a recent fire incident in Putra Heights, Selangor. The system will prompt the user to enter the current gas pressure for each hour of monitoring. The program will continue prompting for pressure readings every 1 hour for 24 hours.

If the pressure exceeds the safe limit of 250 psi, the system will trigger an alert to warn the control center. If the pressure is within the safe range, the program will display a status message indicating normal operation. Write a pseudocode for this situation. [8 marks]

b) Convert this pseudocode into a complete and accurate Flowchart. [5 marks]

```
Start
   Read spending
   If spending >= 1000 then
         Print "Platinum Member"
   Else if spending >= 500 then
         Print "Gold Member"
   Else
         Print "Silver Member"
   End if
Stop
```

c) An IoT Environmental Monitoring System continuously collects barometric atmospheric pressure readings (in hPa) from remote sensors. Construct a **Flowchart** for a program that repeatedly reads pressure values until a sentinel value of `-1` is entered. The system must: [9 marks]

- Validate that each pressure value is positive (> 0).
- Count and display the total number of valid pressure readings.
- Count and display the number of **High Pressure Anomalies** (> 1013.25 hPa) and **Low Pressure Anomalies** (< 1000.00hPa).
- Track the **minimum pressure** and **maximum pressure** recorded among valid inputs.
- Calculate and display the average atmospheric pressure, minimum pressure, maximum pressure, and anomaly counts upon termination.

3

a) Compare **Procedural Programming** and **Object-Oriented Programming** in terms of code reusability. [2 marks]

Aspect	Procedural Programming	Object-Oriented Programming
Code Reusability		

b) Identify the most appropriate programming language paradigm based on the given scenario and give **one (1)** example of the most suitable programming language for each scenario. [4 marks]

Scenario	Type of Programming Language Paradigm	Example of Programming Language
Developer Z is required to create a medical diagnosis system that uses given symptoms and rules to determine possible illnesses.		
Developer Y is assigned to create a temperature conversion tool that follows a step-by-step process and calls functions for calculations.		

c) State the resulting data type from evaluating the following Python expressions:

(i) Recording the exact body temperature of a patient in a hospital system. [1 mark]

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(ii) Storing a user's vehicle registration number plate [1 mark]

_____

d)

(i) Convert the following for statements to while statements. [3 marks]

```
summary = 0
for i in range(12, 45, 7):
      if i % 3 == 0:
          summary += i * 2
      elif i % 2 == 0:
          summary -= i // 2
      else:
          summary += i
print("Summary Total:", summary)
```

(ii) State the output for the coding below. [3 marks]

`sum = 0`
`for i in range(12, 45, 8):`
`  print (i)`
`  sum = sum + i`
`print("Sum for all numbers is :", sum)`	

4

a) State available software for writing Python programs. [1 mark]

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b) Differentiate between a single selection and a multiple selection. [2 marks]

c) Write Python statement for the following algebraic equations. [2 marks]

(i) $y = 3x^5 + 4x^2 - 9$

d) Evaluate the exact value and data type of `ans1` and `ans2` in Python given `x = 17`, `y = 5`, and `z = 2.0`: [2 marks]

(i) `ans1 = x // y + x % y ** z`

(ii) `ans2 = x > y * 3 and not (y <= z or x != 17)`

5

Control structures are specific commands in imperative programming languages that are used to control the logical flow of the program during its execution. Write a Python segment based on a control structure.

a) Protiga Car Sdn Bhd motivates their salesmen by giving incentives at the end of the year. Each salesman will get a 20% yearly bonus based on their total sales of the current year. Write a program segment that can calculate and display the bonus for a salesman based on the total sales inserted.

[3 marks]

b) The following program segment belongs to HiSpeed Sdn Bhd which is a bus company located in Changloon. The program segment calculates the total ticket price based on code and quantity of ticket. An additional travel insurance worth 5% of ticket price will be imposed to each passenger. At the end of the program, the total ticket price is displayed as the final result. Complete the program segment below. [5 marks]

Code	Destination	Price (RM)
100	Kuala Lumpur	60.00
110	Johor Bahru	85.00
111	Kelantan	70.00

6

Write a Python program that applies repetition control (looping). The program must repeatedly ask the user to enter exam marks for several students. The loop will only terminate (sentinel value) when the user enters -1. After the loop ends, the program must calculate and display: [14 marks]

- The total number of students tested.
- The overall average mark of the class.
- The number of students who successfully scored an A (marks 80 and above).
- The number of students who need consultation (marks less than 50).